

Simplified 6-DoF Optical Sensor Mathematical Model

1. Model Assumptions

To reduce the opto-mechanical system to a purely geometric and radiometric model, the following assumptions are made:

1. **Perfect Absorption:** The internal cavity walls are perfectly matte black; stray light, reflections, and multi-path scattering are negligible.
2. **Point Source Emitters:** The LEDs act as ideal point sources with a modified Lambertian emission profile.
3. **Planar Apertures:** The slits above the phototransistors act as infinitesimally small planar apertures with area A , meaning irradiance is uniform across the slit.
4. **Straight-Line Propagation:** Light travels through the air gap without refraction, diffraction, or atmospheric attenuation.

2. Geometry and Coordinate Frames

Let the static bottom plate define the global reference frame with its origin at the center.

The positions of the i -th LED, where $i \in \{1, 2, 3\}$, are defined by their radial distance r_L and angular placement θ_i :

$$\mathbf{p}_{L,i} = \begin{bmatrix} r_L \cos \theta_i \\ r_L \sin \theta_i \\ 0 \end{bmatrix}, \quad \hat{\mathbf{n}}_{L,i} = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} \quad (1)$$

where $\hat{\mathbf{n}}_{L,i}$ represents the static normal vector of the LED emitting surface.

The nominal positions of the j -th slit (where $j \in \{1, 2\}$) on the top plate, separated by a nominal gap distance z_0 , at radial distance r_S and angles γ_j , are:

$$\mathbf{p}_{S,j} = \begin{bmatrix} r_S \cos \gamma_j \\ r_S \sin \gamma_j \\ z_0 \end{bmatrix}, \quad \hat{\mathbf{n}}_{S,j} = \begin{bmatrix} 0 \\ 0 \\ -1 \end{bmatrix} \quad (2)$$

3. 6-DoF Kinematic Transformation

The top plate undergoes a rigid-body transformation described by a translation vector $\mathbf{t} = [t_x, t_y, t_z]^T$ and a rotation matrix $\mathbf{R} \in SO(3)$ representing roll, pitch, and yaw.

Applying this transformation relative to the nominal center of the top plate yields the displaced slit positions $\mathbf{p}'_{S,j}$ and displaced normal vectors $\hat{\mathbf{n}}'_{S,j}$:

$$\mathbf{p}'_{S,j} = \mathbf{R} \left(\mathbf{p}_{S,j} - \begin{bmatrix} 0 \\ 0 \\ z_0 \end{bmatrix} \right) + \begin{bmatrix} 0 \\ 0 \\ z_0 \end{bmatrix} + \mathbf{t} \quad (3)$$

$$\hat{\mathbf{n}}'_{S,j} = \mathbf{R} \hat{\mathbf{n}}_{S,j} \quad (4)$$

4. Radiometric Light Propagation

For a given LED i and slit j , the displacement vector $\mathbf{v}_{i,j}$ from the LED to the slit is:

$$\mathbf{v}_{i,j} = \mathbf{p}'_{S,j} - \mathbf{p}_{L,i} \quad (5)$$

The Euclidean distance $d_{i,j} = \|\mathbf{v}_{i,j}\|$ and the normalized directional vector $\hat{\mathbf{v}}_{i,j} = \mathbf{v}_{i,j}/d_{i,j}$ dictate the optical geometry.

The emission angle θ_{emit} (relative to the LED normal) and the angle of incidence ϕ_{inc} (relative to the slit normal) are determined via the dot product:

$$\cos(\theta_{emit}) = \hat{\mathbf{n}}_{L,i} \cdot \hat{\mathbf{v}}_{i,j} \quad (6)$$

$$\cos(\phi_{inc}) = \hat{\mathbf{n}}'_{S,j} \cdot (-\hat{\mathbf{v}}_{i,j}) \quad (7)$$

The luminous intensity $I(\theta_{emit})$ follows a modified Lambertian profile dictated by the LED's viewing half-angle θ_{half} :

$$I(\theta_{emit}) = I_0 \cos^m(\theta_{emit}), \quad \text{where } m = \frac{-\ln(2)}{\ln(\cos \theta_{half})} \quad (8)$$

where I_0 is the peak axial intensity.

5. Total Irradiance and Transistor Power

Applying the inverse-square law and Lambert's cosine law, the irradiance $E_{i,j}$ arriving at slit j from LED i is:

$$E_{i,j} = \begin{cases} \frac{I_0 \cos^m(\theta_{emit})}{d_{i,j}^2} \cos(\phi_{inc}) & \text{if } \cos(\theta_{emit}) > 0 \text{ and } \cos(\phi_{inc}) > 0 \\ 0 & \text{otherwise} \end{cases} \quad (9)$$

By the principle of superposition, the total radiant power P_j entering the j -th phototransistor through a slit of area A_j is the sum of the contributing irradiances from all three LEDs:

$$P_j = A_j \sum_{i=1}^3 E_{i,j} \quad (10)$$